

Part 1a: Perfect Competition and Monopoly

Instructor: Oleg Baranov

Microeconomics 2 (*Module 1, 2025*).

Suggested Readings for this Part

Readings in Class Textbooks:

► **Varian, 8th edition:**

- Ch 22: Firm Supply
- Ch 23: Industry Supply
- Ch 24: Monopoly

► **Carlton and Perloff:**

- Ch 3: Competition
- Ch 4: Monopolies, Monopsonies and Dominant Firms

Additional Readings on Smart LMS:

- Amazon and Hachette Dispute

Review: Cost Concepts

Cost Concepts - I

- ▶ Denote $F \geq 0$ **fixed costs** (independent of quantity q) and $VC(q) \geq 0$ **variable cost** of producing quantity q
- ▶ Assume that VC is continuous at $q = 0$ and $VC(0) = 0$
- ▶ The total cost of producing quantity q is given by

$$C(q) = F + VC(q)$$

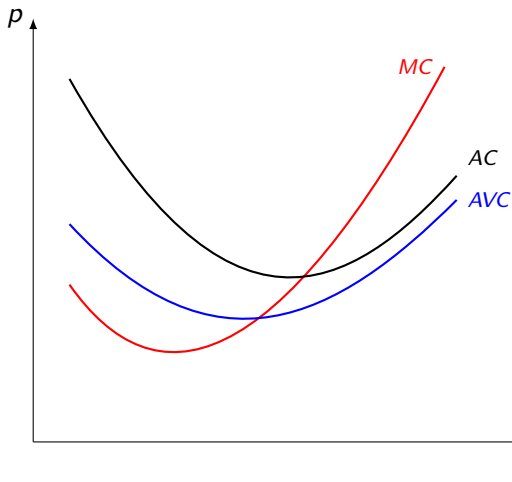
- ▶ Marginal (*incremental*) cost for quantity q is

$$MC(q) = C'(q) = VC'(q)$$

- ▶ Average costs (AC) and average variable costs (AVC) for quantity $q > 0$ are

$$AC(q) = \frac{C(q)}{q} \quad \text{and} \quad AVC(q) = \frac{VC(q)}{q}$$

Cost Diagram



Note that MC curve hits AC and AVC curves at their minimums (interior).

Cost Concepts - II

- ▶ Fixed costs F can be further decomposed into **sunk costs** and **avoidable fixed costs**.
- ▶ The total cost of producing quantity q is given by

$$C(q) = \begin{cases} S & q = 0 \\ F + VC(q) & q > 0 \end{cases}$$

where

- S is the sunk cost part of fixed costs ($S \leq F$)
- $F - S$ is the avoidable part of fixed costs
- **(remark)** teachers sometimes assume that $S = F$

Perfect Competition

Competitive markets

Perfectly competitive market is an **ideal** (a benchmark) against which we compare other models/markets.

Main assumptions of perfect competition:

- ▶ Homogeneous good
- ▶ Perfectly divisible output
- ▶ Perfect information (*Module 2*)
- ▶ No transaction costs
- ▶ No externalities (*Module 2*)
- ▶ **Price-taking behavior (Module 1)**

Perfectly competitive markets are rarely observed in real world (if ever).

Perfect competition

How much does a firm produce?

Firm believes that the market price is not affected by its output choice (i.e., **price-taking behavior**).

- ▶ Profit $\pi(q) = R(q) - C(q) = p \cdot q - C(q)$
- ▶ Firm maximizes its profit $\pi(q)$ by choosing $q \geq 0$

Firm's decision process (a two-step process):

1. What positive output q **maximizes** the profit?
2. Does the firm make enough profit to cover its costs, or is it better off by **shutting down** ($q = 0$)?

Step 1: Maximize profit when $q > 0$

Firm's profit is given by

$$\max_{q>0} p \cdot q - C(q)$$

FOC (interior optimum) — price equals marginal cost at $q^* > 0$:

$$p = MC(q^*)$$

Not done yet! We have to check the SOC:

$$-MC'(q^*) \leq 0 \quad \Rightarrow \quad MC'(q^*) \geq 0$$

Intuitively, marginal costs must be increasing at q^* !

Step 2: Check the shutdown condition

- ▶ We have to check that **shutdown is suboptimal**:

$$\pi(q^*) \geq \pi(0)$$

$$p \geq AVC(q^*) + \frac{F-S}{q^*}$$

or

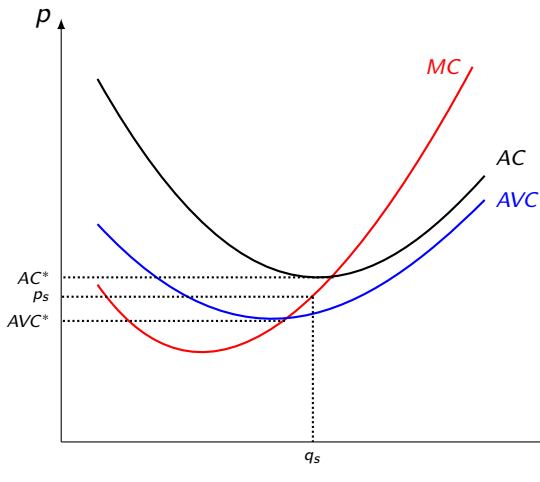
$$MC(q^*) \geq AVC(q^*) + \frac{F-S}{q^*}$$

- ▶ the price is high enough to cover all avoidable costs
 - if $S = F$ (*all fixed costs are sunk*), then $p \geq AVC(q^*)$
 - if $S = 0$ (*no sunk costs*), then $p \geq AC(q^*)$
- ▶ Solve for q_s and p_s using the following equations:

$$MC(q_s) = AVC(q_s) + \frac{F-S}{q_s}$$

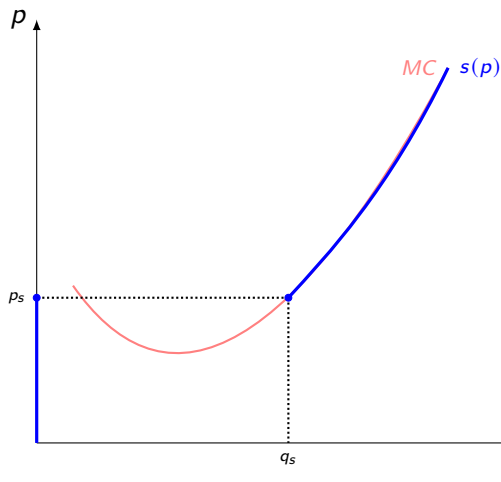
$$p_s = MC(q_s)$$

When to shutdown?



The shutdown price p_s can be as low as AVC^* or as high as AC^* .

Typical Firm's Supply – optimal output given any p



$$s(p) = \begin{cases} 0 & p < p_s \\ 0 \text{ or } q_s & p = p_s \\ q^* & p > p_s \end{cases}$$

where q^* is determined by $MC(q^*) = p$.

Class Example - 1

- Suppose that the firm's total cost of producing q is given by

$$C(q) = \begin{cases} 12 & q = 0 \\ 16 + q^2 & q > 0 \end{cases}$$

- Construct firm's supply function $s(p)$

Short-run Competitive Equilibrium

- ▶ **Demand side:** market demand is given by $D(p)$
- ▶ **Supply side:** N firms with supply curves $s_1(p), \dots, s_N(p)$, then the **market supply** is given by

$$S(p) = \sum_{i=1}^N s_i(p)$$

- ▶ **Market clearing price:** Price p^* that equalizes supply and demand

$$S(p^*) = D(p^*)$$

- ▶ Note that such p might fail to exist
 - ▶ There can be firms with $s_i(p^*) = 0$, $\pi > 0$ or $\pi < 0$.
- ▶ This concept is known as **the short-run equilibrium**.

Class Example - 2

- ▶ Market demand is given by

$$D(p) = 20 - p$$

- ▶ Suppose that all $N \geq 1$ firms are identical.
- ▶ Suppose that a firm's total cost of producing q is given by

$$C(q) = \begin{cases} 12 & q = 0 \\ 16 + q^2 & q > 0 \end{cases}$$

- ▶ Find the short-run equilibrium for all N (if it exists)

Long-run Competitive Equilibrium

- ▶ Suppose that all firms are identical (same costs), there are no sunk costs (i.e., $c(0) = 0$) and entry/exit is free
- ▶ **Main Idea:** # of firms adjusts in the long-run (through entry and exit)
- ▶ **A long-run equilibrium** is a **market price** p^* and a **number of firms** N such that

1. Market supply equals market demand, i.e.

$$N \cdot s(p^*) = D(p^*) , \text{ and}$$

2. Free entry/exit

$$\pi(s(p^*)) = 0$$

This condition is equivalent to $p^* = AC(q^*)$.

- ▶ This concept is known as **the long-run equilibrium**.

Social Welfare

Demand and Inverse Demand Functions

- **From Micro-1:** a consumer with a quasilinear utility for two goods

$$v(q) + m \quad \text{where} \quad v'(q) > 0, \quad v''(q) < 0, \quad v(0) = 0$$

$$\max_{q \geq 0} v(q) - pq \quad \implies \quad \begin{aligned} p &= mv'(q) = p(q) \\ q &= mv^{-1}(p) = D(p) \end{aligned}$$

- The **demand function** $D(p)$ is the quantity the consumer wants to buy at a **per unit price** p .
- The **inverse demand function** is given by

$$p(q) = mv'(q) = D^{-1}(q)$$

and it is **the highest per-unit price** that the consumer is willing to pay for quantity q

- Note that

$$\int_0^q p(x) dx = \int_0^q mv'(x) dx = v(q)$$

Social welfare

- If q units are bought at $p(q)$, then **consumer surplus (CS)**

$$CS := \underbrace{\int_0^q p(x) dx}_{\text{value}} - \underbrace{p(q) \cdot q}_{\text{total payment}}$$

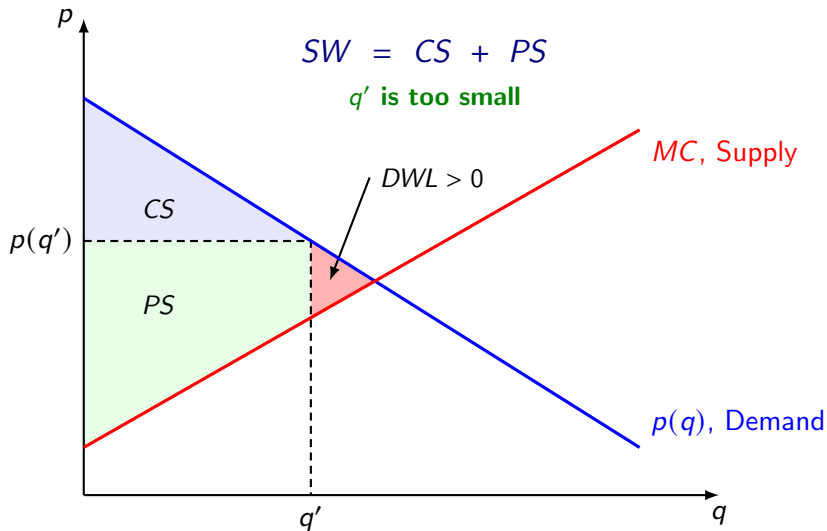
- If q units are sold at $p(q)$, the **producer surplus (PS)**

$$PS(q) := \underbrace{p(q) \cdot q}_{\text{total payment}} - \underbrace{\int_0^q MC(x) dx}_{\text{variable costs}}$$

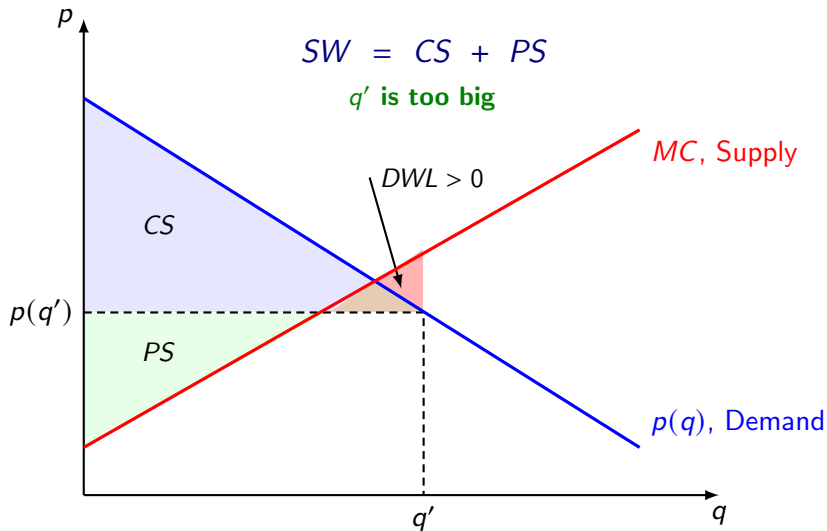
- The sum of CS and PS is called **social welfare (SW)**

$$SW := CS + PS$$

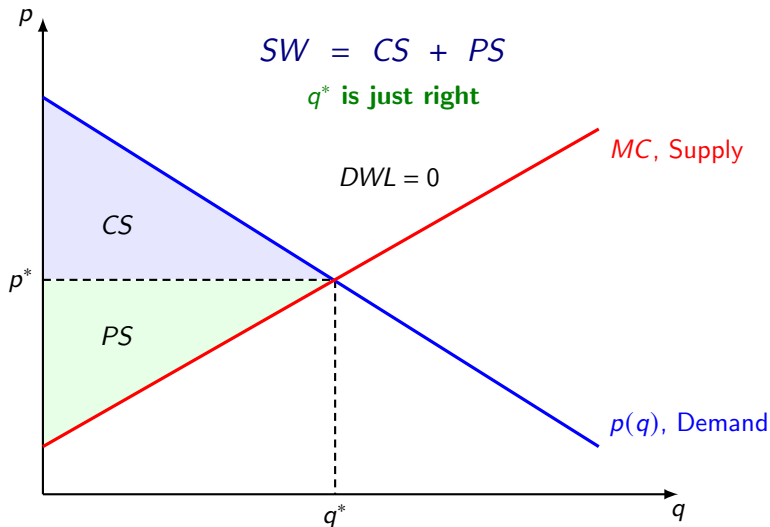
Maximizing Social Welfare



Maximizing Social Welfare



Maximizing Social Welfare



This is why perfect competition is an ideal. It maximizes the social welfare.

Class Example - 3

- ▶ Market demand is given by

$$D(p) = 30 - p$$

- ▶ Suppose that a firm's total cost of producing q is given by

$$C(q) = q^2 \quad \forall q \geq 0$$

- ▶ Calculate CS , PS , DWL and SW in the perfect competition model with a single firm.

Monopoly

Monopoly and its instruments

A firm is a **monopoly** if it is the only seller of a product in a given market.

What can a monopolist do? Many things, including:

- ▶ Post any **per unit price** p (linear pricing).
- ▶ Charge some fixed fee upfront and post a per unit price – **two part tariff** (e.g. *rent a scooter*)
- ▶ Post different prices for different consumers – **3rd degree price discrimination** (e.g. *museums*)
- ▶ Offer different varieties/quantities of the same good at different prices – **2nd degree price discrimination** (e.g., *volume discounts, drinks in bottles/cans*)
- ▶ If several goods are available, then sell them as a package – **bundling** (e.g., *business lunch*)

What leads to a monopolized market?

There are two key drivers:

- ▶ **Technology** – the average cost curve and the minimum efficient scale (AC^*) in the long-run
- ▶ **The size of the market** — demand curve

If only 1-2 firms can potentially coexist in the market in the long run due to

- ▶ high AC^* , and
- ▶ small market (small relative to AC^*)

we should expect monopolies to appear.

Monopoly: Linear Pricing

Linear pricing

- ▶ Suppose that a monopolist charges every consumer the **same per unit price** p . This is called **linear pricing**.
- ▶ The total sales are then given by the market demand

$$q = D(p)$$

- ▶ The profit of the monopolist is then given by

$$\pi(p) = p \cdot D(p) - C(D(p))$$

- ▶ What price should the monopolist set?

$$\max_{p \geq 0} \pi(p)$$

- ▶ We will solve an **equivalent** problem $\max_{q \geq 0} \pi(q)$

Profit Maximization (quantity)

- ▶ The inverse demand function is given by $p(q) = D^{-1}(q)$.
- ▶ The monopolist revenue is given by

$$R(q) = p \cdot D(p) = p(q) \cdot q$$

- ▶ Then **marginal revenue** (abbreviated as MR) is as follows:

$$MR(q) = R'(q) = p(q) + p'(q) \cdot q$$

- ▶ The monopolist chooses output level to maximize its profit

$$\max_{q \geq 0} \pi(q) = R(q) - C(q)$$

- ▶ Monopolist's decision process:
 - ▶ What positive output q **maximizes** the profit?
 - ▶ Does the firm make enough profit to cover its costs, or it is better off by **shutting down** ($q = 0$)?

Solution to Profit Maximization

FOC (interior optimum) — marginal revenue equals marginal cost:

$$MR(q^m) = MC(q^m)$$

We are not done yet!

1. Need to check SOC

$$MR'(q^m) - MC'(q^m) \leq 0$$

2. Have to check that shutdown is suboptimal

$$\pi(q^m) \geq \pi(0)$$

$$p^m \geq AVC(q^m) + \frac{F-S}{q^m}$$

$$MC(q^m) - p'(q^m)q^m \geq AVC(q^m) + \frac{F-S}{q^m}$$

This inequality is easier to satisfy in compar. with the PC scenario.

Class Example - 4

- ▶ Inverse demand function is given by

$$p(q) = K - q \quad \text{where } K > 0$$

- ▶ Suppose that the firm's total cost of producing q is given by

$$C(q) = \begin{cases} 12 & q = 0 \\ 16 + q^2 & q > 0 \end{cases}$$

- ▶ Find the optimal monopoly quantity-price pair for $K = 4$ and $K = 8$.

Reminder: Price Elasticity of Demand

For (p, q) on the demand curve, price elasticity is given by

$$\epsilon(p) = \frac{p \cdot D'(p)}{D(p)} \quad \text{using the demand function}$$

and (since $p'(q) = 1/D'(p(q))$)

$$\epsilon(q) = \frac{p(q)}{p'(q) \cdot q} \quad \text{using the inverse demand function}$$

- ▶ percentage change in quantity given a 1% change in price
- ▶ demand is a nonincreasing function, so elasticity is negative
- ▶ $|\epsilon| < 1$ – inelastic demand (*small change in quantity*)
- ▶ $|\epsilon| > 1$ – elastic demand (*large change in quantity*)

Lerner's Index of Market Power (LIMP)

$$\frac{p - MC(q)}{p}$$

- ▶ $p - MC(q)$ is the price-cost margin
- ▶ $\frac{p - MC(q)}{p}$ is the margin as a share of the price
- ▶ the LIMP takes values in $[0, 1]$ interval. For example,
 - ▶ if $p = 10$ and $MC(q) = 9$, then $LIMP = 0.1$
 - ▶ if $p = 10$ and $MC(q) = 1$, then $LIMP = 0.9$
- ▶ the higher the LIMP value, the larger the market power
- ▶ Note that the index of a competitive firm is **zero**

Reinterpreting Monopoly's FOC

FOC (interior optimum) — marginal rev. equals marginal costs:

$$\begin{aligned}MR(q^m) &= MC(q^m) \\p^m + p'(q^m) q^m &= MC(q^m) \\ \frac{p^m - MC(q^m)}{p^m} &= \frac{-p'(q^m) q^m}{p^m} \\ \frac{p^m - MC(q^m)}{p^m} &= -\frac{1}{\epsilon(q^m)}\end{aligned}$$

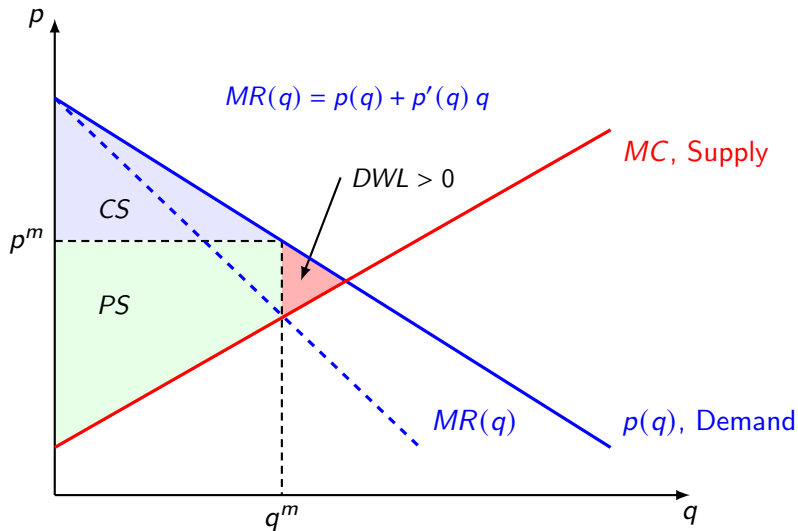
- ▶ Left-hand side is the Lerner's Index of Market Power (LIMP).
- ▶ Right-hand side is the inverse of demand elasticity
- ▶ For example:

if $\epsilon = -11$, then $p = 1.1MC$ and $LIMP = 0.09$.

if $\epsilon = -1.1$, then $p = 11MC$ and $LIMP = 0.91$.

- ▶ Note that monopoly never operates in the inelastic part of the demand curve (i.e., $-1 < \epsilon < 0$). Why?

Graphical Analysis of Social Welfare



Since $MR(q) < p(q)$, there is always undersupply $q^m < q^*$ and positive DWL .

Social costs

Why monopoly is bad?

- ▶ The higher the price, the lower the **consumer surplus CS** .
- ▶ In general, a monopoly is inefficient resulting in a **deadweight loss DWL** .
- ▶ The total welfare loss can be even higher than DWL due to **rent-seeking** (profits can be used to gain and protect the market power) *e.g. lobbying, advertising*.

Government often regulates and limits market power:

- ▶ Price caps — oil and gas prices.
- ▶ Limitations on mergers (consolidation) — need to get an approval from the government.

Social benefits

Can monopoly be good?

- ▶ Incentives to innovate (*new products, more efficient production*), patents (*protects R&D*).
- ▶ Sometimes a monopoly may be the most efficient way to produce: *postal service, subway*. This situation is called a **natural monopoly**.
- ▶ A natural monopoly may exist if the industry cost function is **subadditive**:

$$C(q_1 + q_2) \leq C(q_1) + C(q_2) \quad \forall q_1, q_2 \geq 0$$

- ▶ This is very common for industries with large fixed costs (e.g., public utilities)

Monopsony

Monopsony

A firm is a **monopsony** if it is the only buyer of a product in a given market — “Nornikel” hiring labor in Norilsk.

Monopsony is a flipside of the monopoly:

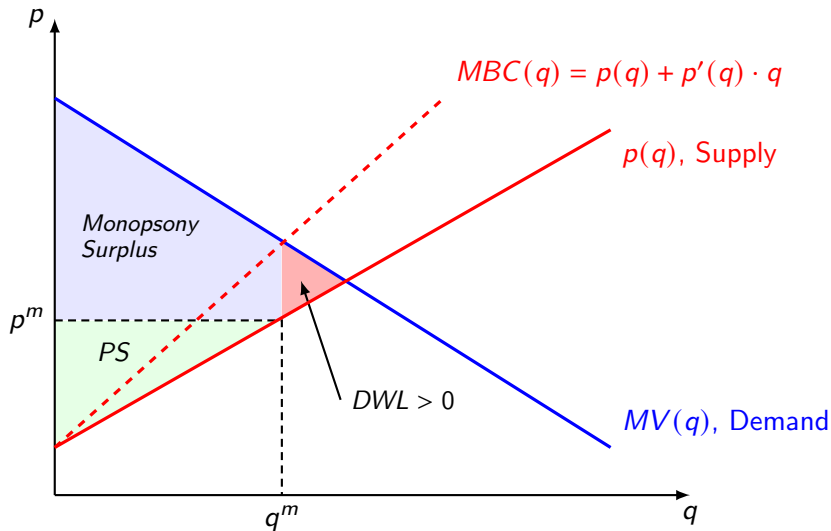
- ▶ Value of buying quantity q : $V(q)$ (*MV for marginal*)
- ▶ $p(q)$ is the **inverse supply**.
- ▶ Cost of buying q units: $BC(q) = p(q) \cdot q$
- ▶ Marginal cost of buying q units

$$MBC(q) = p(q) + p'(q) \cdot q$$

- ▶ Monopsony's profit: $\pi(q) = V(q) - BC(q)$
- ▶ Monopsony will chose q^m to satisfy

$$MV(q^m) = MBC(q^m)$$

Graphical Analysis



Since $MBC(q) > p(q)$, there is always undersupply $q^m < q^*$ and positive DWL.

Case Study: Amazon

Background

Amazon was founded in 1986 as an online book retailer.

Currently Amazon is the largest online retailer.

Issue: Dispute with **Hachette** over cut of supply prices for books

- ▶ Amazon wanted a larger share.
- ▶ Hachette refused.
- ▶ Amazon began disrupting sales of Hachette books:
 - Delay deliveries.
 - Steer away customers.
 - Raise prices.
 - Prevent pre-orders.

Market power

- ▶ Amazon is the largest seller, but it does not use its **monopoly power** — keeps final prices low for its customers. **Why?**
- ▶ Amazon is also a **monopsony** as it is the largest buyer of books from publishers.
- ▶ Amazon is abusing its **monopsony power** by driving supply prices down

“Amazon has too much power and it uses that power in ways to hurt America” - Paul Krugman

How to solve problems with multiple consumers?

Step 1: Demand Aggregation

- ▶ Demand function $D(p)$ tells us the quantity that a consumer wants to buy at **per unit price** p .
- ▶ Inverse demand function tells the highest per-unit price p that the consumer is willing to pay and still buy q units
- ▶ If we have N consumers with demand functions $D_1, \dots, D_N$, then the **market demand** is given by

$$D(p) = \sum_{i=1}^N D_i(p)$$

- ▶ The market demand function itself can be inverted, i.e., $p(q) = D^{-1}(q)$. This is **inverse market demand**.
- ▶ In general,

$$p(q) \neq \sum_{i=1}^N p_i(q)$$

Step 2: Finding monopoly outcome

- ▶ Suppose that there are N consumers with individual demands $D_i(p)$ and inverses $p_i(q)$.
- ▶ Market demand and inverse market demand are given by

$$D(p) = \sum_{i=1}^N D_i(p) \quad \Rightarrow \quad p(q) = D^{-1}(q)$$

- ▶ The optimal quantity q^m and per unit price p^m are given by

$$MR(q^m) = MC(q^m) \quad \Rightarrow \quad p^m = p(q^m)$$

- ▶ Individual quantities bought by consumers are given by

$$q_i = D_i(p^m)$$

Skill checklist for Part 1a

1. Find a market demand/inverse demand.
2. Find a market supply with N competitive firms.
3. Find a competitive equilibrium: price, each consumer's/firm's quantity.
4. Determine a number of competitive firms with free/costly entry.
5. Find a monopoly/monopsony uniform price and each consumer's quantity (*FOC, SOC, shutdown condition*).
6. Compute CS, PS, SW, DWL in each case.